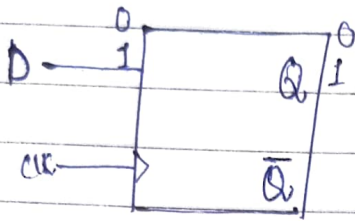


D Q_{n+1}
 0 0
 1 1

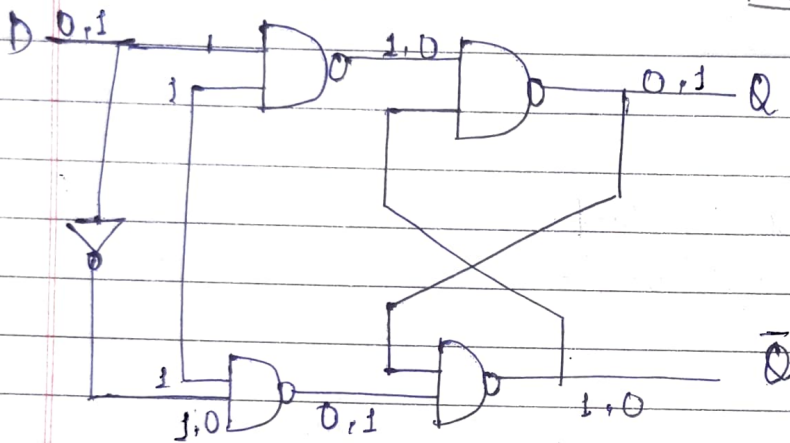
D flip-flop $\rightarrow$ Transparent latch



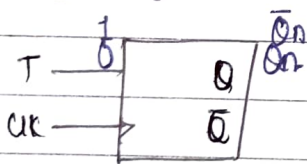
(No matter)

D	Q	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

$Q_{n+1} = D \rightarrow$ char. eqn



T flip-flop



Truth table

T	Q_{n+1}
0	$Q_n \rightarrow$ No action
1	$\bar{Q}_n$

Char. table

T	Q_n (prev. o/p)	Q_{n+1} (next o/p)
0	0	0
0	1	1
1	0	1
1	1	0

Excitation table

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

$Q_{n+1} = T \oplus Q_n$

One's Complement

1 0 1 0 1 1
0 1 0 1 0 1

$$\begin{array}{r} 101011 \\ - 010100 \rightarrow 1's \\ \hline + 1 \rightarrow 2's \\ \hline 010101 \end{array}$$

$$\begin{array}{r} 10100 \\ 01011 \rightarrow 1's \\ 10101 \rightarrow 2's \\ 01100 \rightarrow 2's \end{array}$$

Subtraction using 1's comp.

Ques. $(101011)_2 - (111001)_2$

$$\begin{array}{r} 101011 \\ + 000110 \\ \hline 110001 \end{array}$$

1's comp. $(110001) \rightarrow$ if there is no carry $\rightarrow$ no is -ve

$$\begin{array}{r} 101011 \\ + 000110 \\ \hline 110001 \\ \hline 001110 \rightarrow \text{final ans} \end{array}$$

1's

Ques. $(110101)_2 - (100101)_2$

$$\begin{array}{r} 110101 \\ + 011010 \\ \hline 1001111 \end{array}$$

if there is carry $\rightarrow$ no is +ve

$$\begin{array}{r} 1001111 \\ + 1 \\ \hline 0100000 \rightarrow \text{final ans.} \end{array}$$

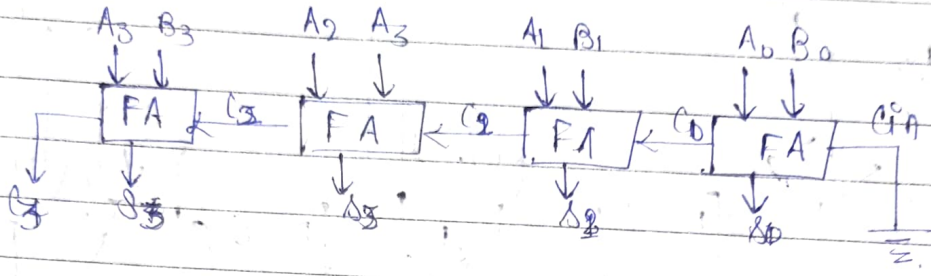
carry

Que. Design a digital circuit when perform 9+5

9 → 1001

5 → 0101

1110

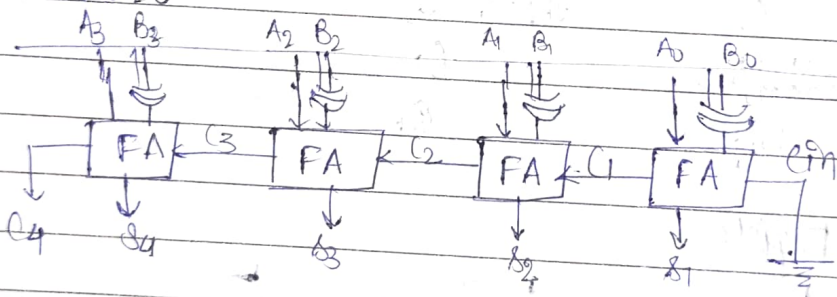


Que. Design a digital circuit to perform 9-5

9 → 1001

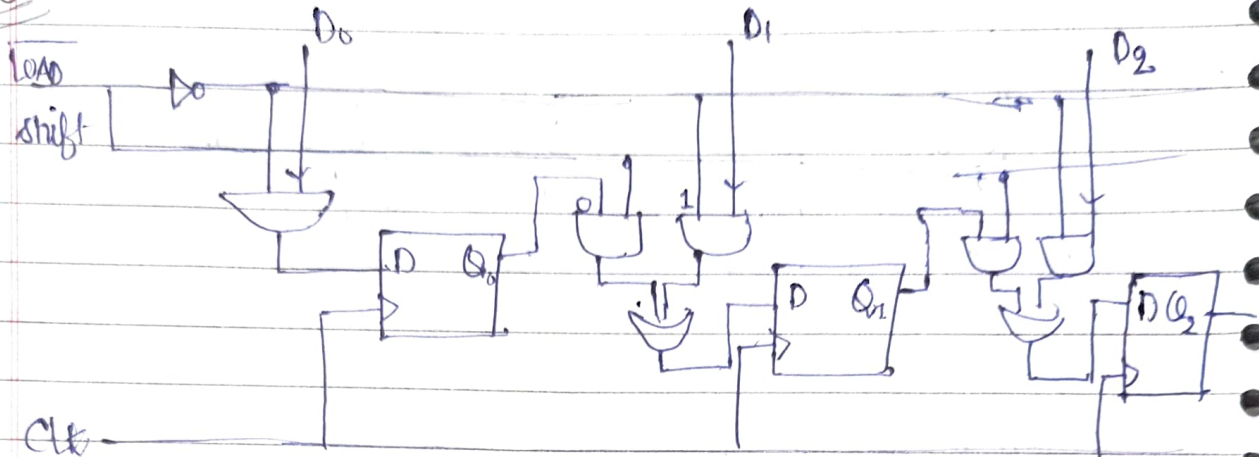
5 → 0101

-5 → 1010



flip-flop - store 1 bit either logic 0 or logic 1
A set of registers form a memory

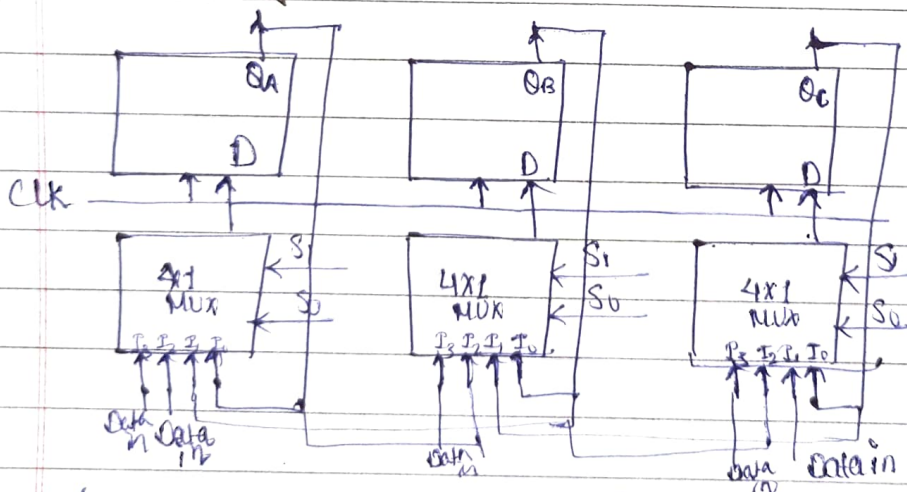
PISO



bit → shift right type register

clk	load	D Input	Q2	Q1	Q0
↑	0				
↑	shift 1				
↑	shift 1				

Universal register



w_1	w_0	Dir	Output
0	0	I_0	Q_A, Q_B, Q_C [No change]
0	1		Q_B, Q_C , new bit shift left
1	0		new, Q_A, Q_B shift right
1	1		new, new, new Parallel load

special type of register

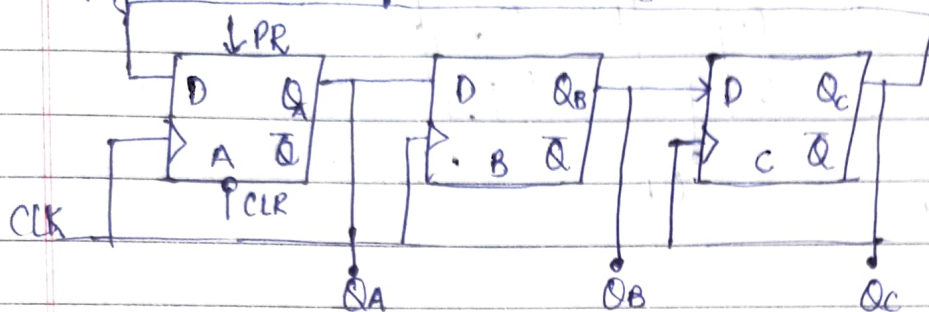
used to repeat bits after certain position

Ring Counter (3-bit)

(Preset)

(Clear)

Asynchronous Flip → PR & CLR



PR → preset is used to set the flip-flop, it can be apply at any position of flip-flop.

At preset, flip-flop set to '1'. set output at '1'.

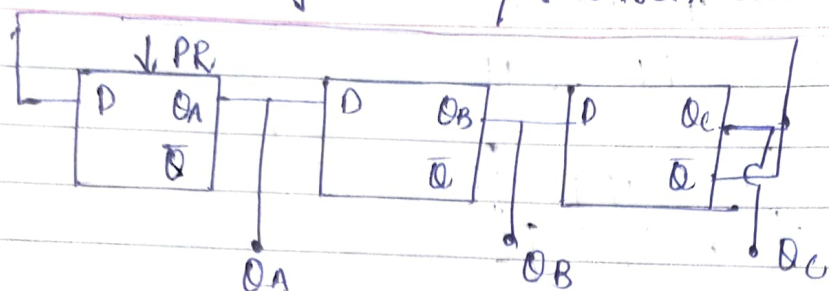
CLR → Reset, set output at '0'.

preset & reset are independent to clk.

clk	Q _A	Q _B	Q _C	Decimal no
1.	0	0	0	0
2.	1	0	0	4
3.	0	1	0	2
4.	0	0	1	1
5.	1	0	0	4
6.	0	1	0	2
7.	0	0	1	1

Twisted Ring counter / Johnson counter

DATE _____
PAGE _____



CLK	QA	QB	QC	
1	0	0	0	0
2	1	0	0	4
3	1	1	0	6
4	1	1	1	7
5	0	1	1	3
6	0	0	1	1
7	0	0	0	0
8	1	0	0	4

* Counter :- It is used to count no. of clock signals or pulses

→ Two types

- up counter $\rightarrow 0, 1, 2, \dots, n$
- down counter $\rightarrow n, n-1, n-2, \dots, 0$

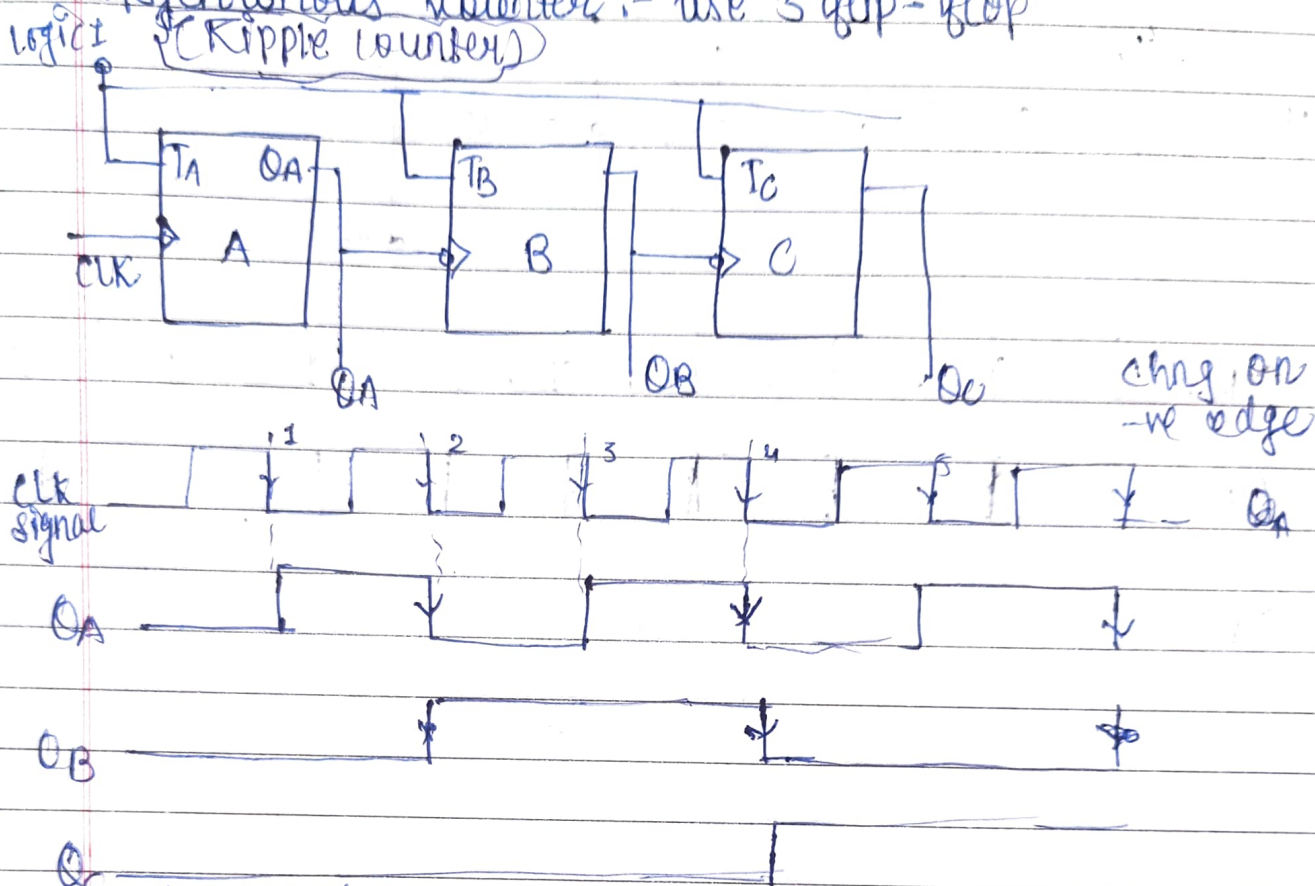
Asynchronous counter :- apply diff. freq. clock on every flip-flop
Synchronous counter :- if apply same freq. clock on every flip-flop

Sequential $\Rightarrow$ flip-flop, counter [$N \leq 2^n$]

No. of count state

no. of flip-flop

Asynchronous counter :- use 3 flip-flop
(Ripple counter)



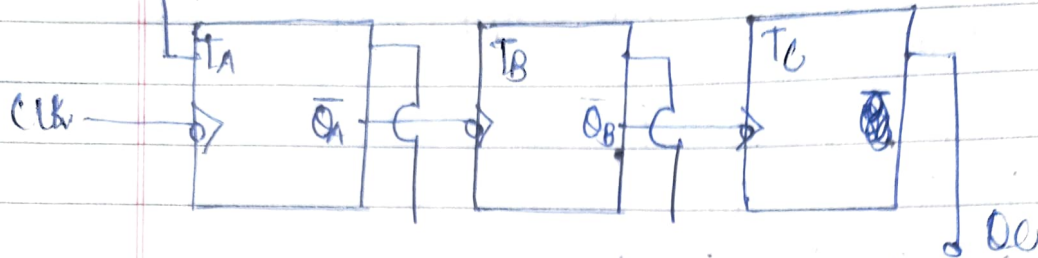
up counter

clk	QA	QB	QC	Decimal
0	0	0	0	0
1	0	0	1	1
2	0	1	0	2
3	0	1	1	3
4	1	0	0	4
5	1	0	1	5
6	1	1	0	6
7	1	1	1	7

logic 1

Down counter

DATE
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QB

QC

CLK	QC	QB	QA
0	0	0	0
1			
2			
3			
4			
5			
6			
7			

up-down counter

